

Case Study 1

This report aims to understand whether or not a team's franchise value can be better explained by the number of their Facebook and Twitter followers. The following dataset from Statista (<https://www.statista.com/topics/963/national-football-league/>) in 2024 showcases 32 professional NFL teams...

1. Team name
2. Franchise value in 2024 (billions of U.S dollars)
3. Number of Facebook fans as of November 2024 (millions)
4. Number of Twitter followers as of November 2024 (millions)

Statistical analysis aims to identify patterns and trends from data by collecting and interpreting information.

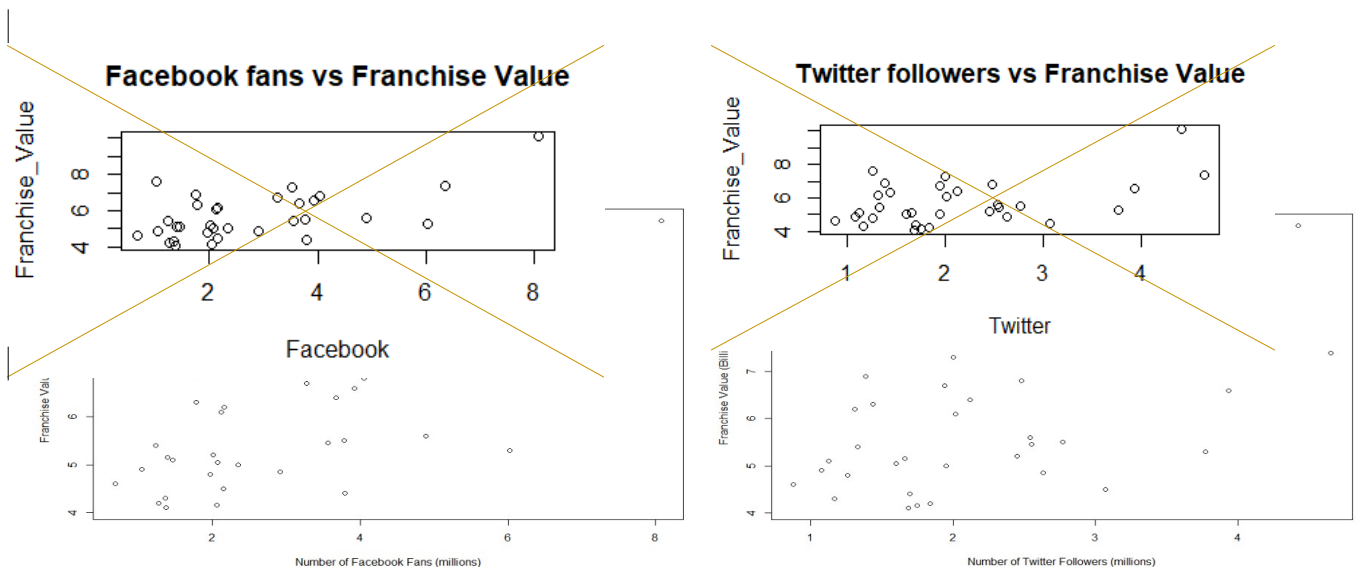
Correlation Analysis:

There seems to be a positive linear correlation between a team's franchise value and the number of their fans/ followers on Facebook/ Twitter, so a larger following on Twitter and Facebook could lead to a larger franchise value. Every value will be rounded 3 decimal points!

Facebook cor value = 0.595

Twitter cor value = 0.464

Exploratory Analysis:



As shown in the images above, there seems to be a correlation in the extremes (super large following = big franchise value), however, that trend seems to be much weaker when looking at the opposite end of the spectrum of the relationship between social media following and franchise value (less following, less value). As we can see, some teams have a greater franchise value than those with a much higher following on both Twitter and Facebook.

As shown in the images above, there is a positive linear relationship between the number of Twitter followers and Franchise Value; The number of Facebook fans and Franchise value also appear to have a positive linear relationship, meaning that on average, teams with more Twitter followers or Facebook fans will most likely have a higher franchise value. Another important thing to notice from these graphs would be the large variability of the Franchise value given the number of Twitter followers/ Facebook fans, the variability seems to increase near the high end of Twitter followers and Facebook fans.

Model Specification 1:

Independent Variable: Number of Facebook Fans

Dependant Variable: Franchise Value

The purpose of Model 1 is to use the number of Facebook fans to explain the franchise value, we can see this as a linear regression equation such as

$$FV = B_0 + B_1(FBfans) + E$$

With FV being our dependent variable, B_0 being our intercept, and B_1 being the effect of fans on franchise value. E is just to account for error.

Model 1 Estimation:

```
call:
lm(formula = Franchise_value ~ Facebook, data =
fanData)

Coefficients:
(Intercept)      Facebook
    4.4385         0.4481
```

Here is the model with estimated coefficients from the data set

FV = 4.4396 + 0.448(#Facebook Fans) (with margin: 0)

This means that if a team were to have 0 Facebook fans then their franchise value would be around 4.436 billion dollars, and for each 1 million Facebook fans a team has, their franchise value increases by about 0.448 billion dollars.

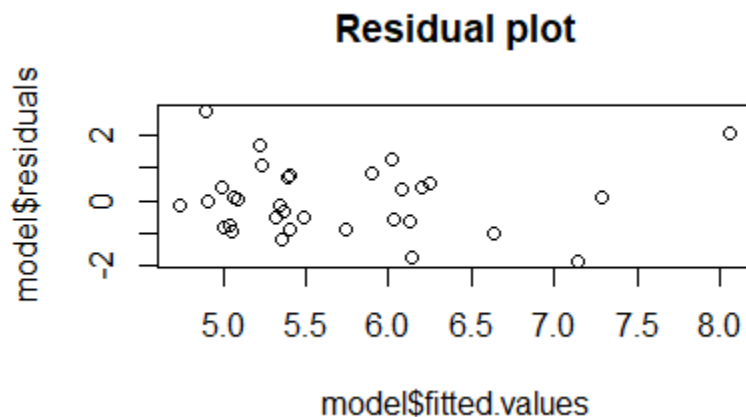
Now we can make a hypothesis test

H0: $B_1 = 0$

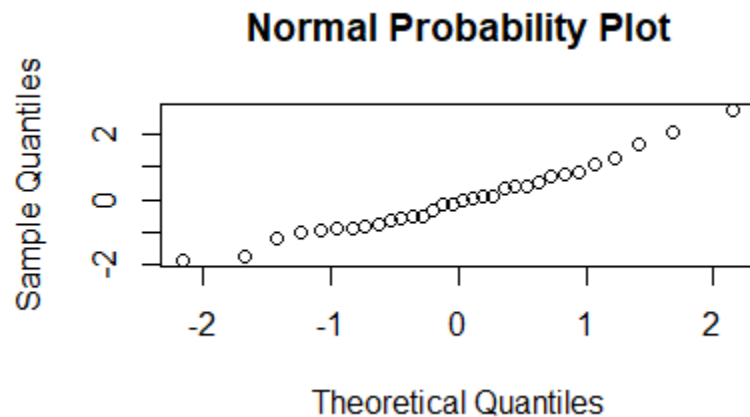
H1: $B_1 \neq 0$

We get a P value of 0.0003 which is $< .05$ meaning that we reject the null hypothesis that the coefficient on Facebook fans is 0 and conclude that there is a significant linear relationship between Facebook fans and franchise value.

Diagnostic Checking 1:



The residuals seem to be centered around zero across fitted values. There are mild changes in variance across fitted values, which satisfies the assumptions of the model. This is also strong evidence that the residuals are not correlated with each other since this plot appears to be random and without any patterns.



Creating a norm Probability lot reveals a generally straight line supporting that the residuals follow a normal distribution

After conducting a Shapiro-Wilk test, we get a p-value of 0.564 which is $> .05$, meaning that we fail to reject the null hypothesis that the residuals follow a normal distribution.



When creating a histogram of the model, it is clear that the data is skewed to the right. However, since we have 3 pieces of evidence in support that the model is a normal distribution, the assumption can be held that the residuals are normally distributed.

Model 1 Assessment:

The r^2 value of Model 1 is .354, meaning that 35.4% of the variability in a team's franchise value can be explained by the number of Facebook fans they have.

Model Specification 2:

Independent Variable: Number of twitter followers

Dependant Variable: Franchise Value

$$FV = B_0 + B_1(Tfans) + E$$

FV: Franchise Value

B₀: Our intercept, the baseline franchise value when there are 0 Twitter followers

TFans: Number of Twitter fans

E: Margin of error

Model Estimation 2:

$$FV = 4.376 + 0.618(TFans)$$

With margin: 0

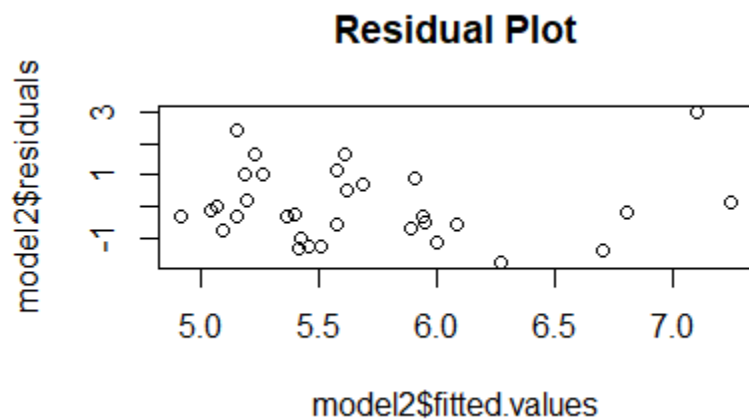
This means that if a team was to have 0 Twitter followers, then their baseline Franchise value would be 4.376 billion dollars, and for every 1 million Twitter followers they have, we can expect an increase of 0.618 billion to the team's franchise value.

$$H_0: B_1 = 0$$

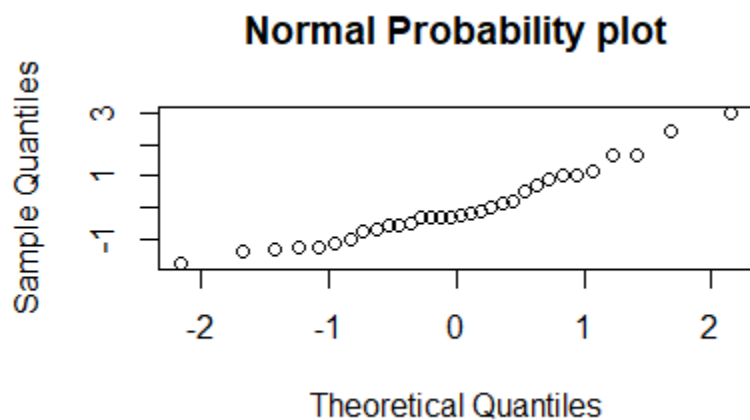
$$H_1: B_1 \neq 0$$

We get a P value of .007 which is $< .05$, meaning that we reject the null hypothesis that the coefficient on Twitter followers is 0 and conclude that there is a significant linear relationship between Twitter followers and franchise value.

Diagnostic Checking 2:



The residuals seem to be centered around zero across fitted values. There are mild changes in variance across fitted values, which satisfies the assumptions of the model. This plot also serves as evidence that the residuals are not correlated with each other due to the lack of a pattern.



After creating a Norm Prob plot, we can see that the residuals, although seem to follow a generally straight line, also have some deviation which would not support the residuals following a normal distribution. residuals follow a generally straight line supporting that they are normally distributed.

A Shapiro-Wilk test gets us a P-value of 0.096 which is > 0.05 , meaning that we fail to reject the null hypothesis that the residuals follow a normal distribution



When creating a histogram of the model, it is clear that the data is skewed to the right. However, since we have 3 pieces of evidence in favor of a normal distribution, we can assume that model 2's residuals follow a normal distribution

Model Assessment 2:

The r^2 value is 0.215 meaning that 21.5% of the variability in a team's franchise value can be explained by the number of Twitter followers they have.

Model Deployment:

Based on my analysis I would recommend Model 2 over Model 1 since the coefficient for Twitter followers is higher than the coefficient for Facebook fans, revealing that Twitter followers can better explain a team's franchise value compared to Facebook fans.

The Kansas City Chiefs franchise value in 2024 is 4.85 billion. My model predicts the franchise value to be 6.002 billion dollars, overestimating the Chiefs franchise value by about 1.152 billion dollars

I would recommend using the number of Facebook fans to predict franchise value (Model 1). Even though both models satisfied the diagnostic checks, model 1 can explain 33.2% of the variability in the franchise value while Model 2 can only explain 21.5% of the variability, meaning that model 1 is a much better predictor for franchise value compared to model 2.

```
{r}
newModel1Data = data.frame(Facebook = c(2.91))
predict(model1, newModel1Data)
```

1
5.742514

With this in mind, we can predict the franchise value of the Kansas City Chiefs, who have 2.91 million Facebook Fans, and after plugging the Kansas City Chiefs' number of fans into our model, we predict that their Franchise value is \$5.743 billion dollars.

	Team	Franchise_Value	Facebook
13	Green Bay Packers	5.00	4.09
14	Denver Broncos	5.50	3.78
15	Seattle Seahawks	5.45	3.56
16	Tampa Bay Buccaneers	5.40	1.23
17	Pittsburgh Steelers	5.30	6.02
18	Atlanta Falcons	5.20	2.01
19	Cleveland Browns	5.15	1.39
20	Los Angeles Chargers	5.10	1.46
21	Minnesota Vikings	5.05	2.07
22	Baltimore Ravens	5.00	2.35
23	Tennessee Titans	4.90	1.04
24	Kansas City Chiefs	4.85	2.91

However, according to our dataset, the Kansas City Chiefs franchise value in 2024 is \$4.85 billion, meaning that the model overestimates the Kansas City Chiefs franchise value by almost \$890 million.